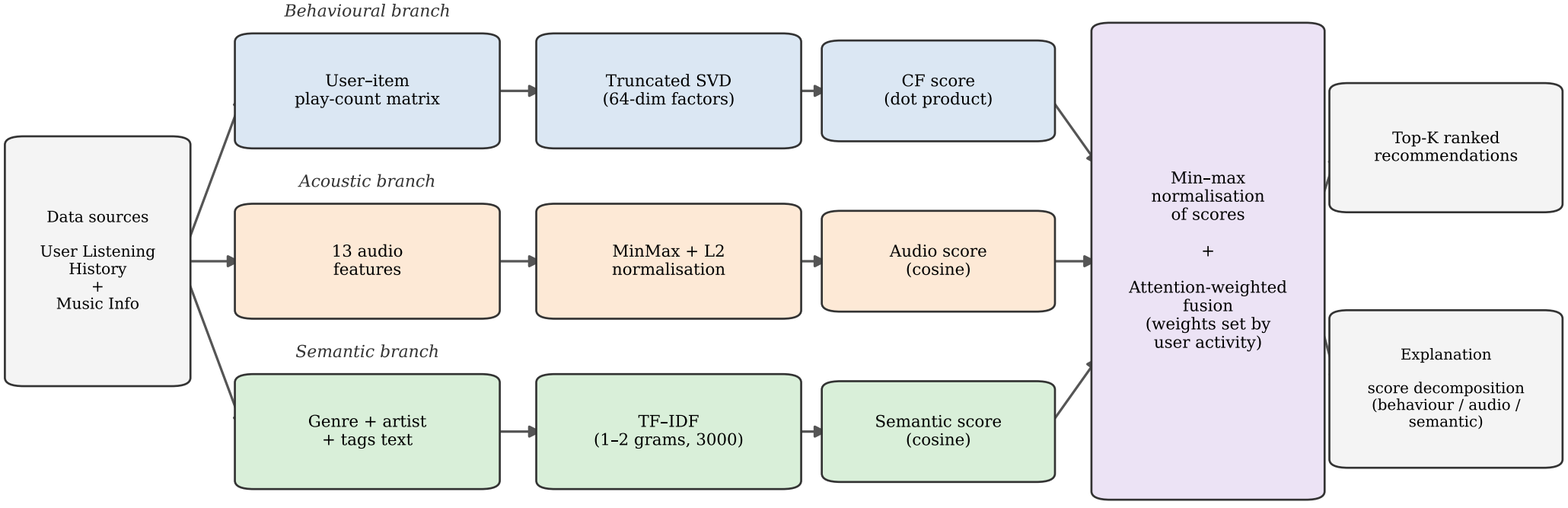
Data sources
User Listening History
+
Music Info
Behavioural branch
User–item play-count matrix
Truncated SVD (64-dim factors)
CF score (dot product)
Acoustic branch
13 audio features
MinMax + L2 normalisation
Audio score (cosine)
Semantic branch
Genre + artist + tags text
TF–IDF (1–2 grams, 3000)
Semantic score (cosine)
Min–max normalisation of scores
+
Attention-weighted fusion (weights set by user activity)
Top-K ranked recommendations
Explanation
score decomposition (behaviour / audio / semantic)